

★ What is a Ratio ?

Definition of Ratio

Ratio is the comparison of two or more numbers in terms of 'how many times'.

In certain situations, the comparison of two quantities by the method of division is very efficient. We can say that the comparison or simplified form of two quantities of the same kind is referred to as ratio. This relation gives us how many times one quantity is equal to the other quantity. In simple words, the ratio is the number which can be used to express one quantity as a fraction of the other ones.

The two numbers in a ratio can only be compared when they have the same unit. We make use of ratios to compare two things. **The sign used to denote a ratio is ':'.**



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What is a Ratio?

(min 2 term)

$$b = 2a$$

$$\frac{a}{b} = \frac{1}{2}$$

$$a:b = 1:2$$

$$a:b = 1:2$$

$$2:4 \quad a:b:c = 1:2:3$$

$$2:4:6$$

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$$a = \frac{b}{2} \checkmark$$

$$a:b = \frac{1}{2}$$

$$\frac{a}{b} = \frac{1}{2}$$

$$(b = 2a)$$

$$d_1:d_2 \checkmark$$

$$s_1:s_2 \checkmark$$

$$t_1:t_2 \checkmark$$

$$S_1:d_2 \times$$

$$w_1:w_2$$

$$\downarrow \quad \downarrow$$
$$kg \quad kg$$

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★ What is a Ratio ?

A ratio can be written as a fraction, say $\frac{2}{5}$, or can be represented by using "to", as "2 to 5" or as $2:5$. We happen to see various comparisons or say ratios in our daily life.



Ratio is denoted as " $a:b$ " or " a/b " or "a to b" where 'a' is called the first term or numerator or antecedent and b is called the second term or denominator or consequent.

Comparison of same item must be done in ratio. A Kilogram can't be compared with a gram. Similarly you cannot compare money with weights.



For Example: Popatlal purchased dozen of lemon worth Rs. 20 and dozen of apples worth Rs. 100. Now suppose we need to compare the cost of apples and lemon.

Therefore, Cost of Apples/ Cost of Lemons = $100/20 = 5$

Hence we can conclude that apples are 5 times more expensive than lemon.





***My Self Respect**

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$$\frac{\text{Cost (Apple)}}{\text{Cost (Lemon)}} = \frac{100}{20} = 5$$
$$\text{Cost (Apple)} = 5 \text{ Cost (Lemon)}$$

What is Proportion ?

A proportion is an equation that says that two ratios are equivalent. For instance if one package of cookie mix results in 20 cookies than that would be the same as to say that two packages will result in 40 cookies. Proportion is denoted by double colon symbol ::

$$20/1=40/2$$

$$y, w \neq 0$$

$$\frac{x}{y} = \frac{z}{w}$$

$$\underline{x:y} :: \underline{z:w}$$

A proportion is read as "x is to y as z is to w"

$$x/y = z/w \text{ where } y, w \neq 0$$

What is Proportion ?

A proportion is read as "x is to y as z is to w" $x/y = z/w$ where $y, w \neq 0$

Here we can say that x, y, z and w are in proportion and it is represented as

$x:y :: z:w$

$x:y :: z:w$

$\frac{x}{y} = \frac{z}{w}$

x, y, z, w

x and w are extreme proportion and y and z are mean proportion.

x is first proportion.

y is second proportion.

z is third proportion.

w is fourth proportion.